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Mother–infant separation and the nature of inter-individual relationships: experiments with rhesus monkeys

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(*Based on a Review Lecture delivered on* 17 *April* 1975 – *TS received* 28 *July* 1976)

Introduction

During the late fifties the extent to which a period of separation from the mother-figure could have adverse effects on the development of a human child was a matter of some dispute. While some psychiatrists and paediatricians held that the consequences could be severe (see for example, Bowlby 1951, 1958; Ainsworth 1962), their views were based largely on clinical and retrospective evidence, and conflicted with established medical and social practice. Since controlled experiments with human subjects were out of the question on ethical grounds, we decided to see what progress could be made with rhesus monkeys. It was apparent from the start that the consequences of a separation experience could not be studied in isolation – it was, for instance, necessary for us to do some initial work on processes of intraspecific communication in rhesus monkeys, and on how the mother–infant relationship develops. Some of this work was reviewed previously (Hinde & Spencer-Booth 1968). As the work progressed it became apparent that (*a*) the consequences of a separation experience vary with the nature of the mother–infant relationship, and (*b*) that the nature of the mother–infant relationship varies with the social situation. Furthermore, a number of conceptual issues arose in the course of the work, such as precisely what one means by a 'relationship', and how the important but somewhat intangible 'natures' or 'qualities' of relationships could be assessed and compared in a hard-headed manner. The aim of this paper is not to review the current state of the evidence bearing on any one of these problems, but rather to demonstrate, from the work of one laboratory, how the various problems are inter-related.

Materials and methods

The work discussed here was carried out with rhesus monkeys nearly all of which were obtained from zoos or other sources in this country, or were bred in the colony.

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The monkeys live in small groups each consisting of a male, three to four females and their young. Each group lives in an outdoor cage (548 cm × 243 cm × 246 cm) communicating with an inside room (185 cm × 133 cm × 239 cm). This situation provides the monkeys with a moderately complex social situation, though one much simpler than that of a natural troop, and at the same time permits a moderate degree of experimental control and moderately precise recording. Until recently, all records were made on check sheets using half minute time intervals (Hinde 1973). More recently a keyboard event recorder system has been used (White 1971); each of the keys on the 39-key keyboard is assigned to a particular aspect of behaviour, and the observer depresses the appropriate keys as he watches. The record is stored initially on magnetic tape and can subsequently be computer analysed and stored on computer magnetic tape.

All the data were collected between 09h00 and 13h00, except that some watches during separation were continued until 14h30 local time.

Development of the mother–infant relationship

At birth, the infant monkey has a limited repertoire of movement patterns which are elicited appropriately by stimuli from the mother. With some help from the latter he clings to her ventrum and is able to find a nipple with his mouth. The subsequent development of the mother–infant relationship involves an ongoing interaction between mother and infant, in which the behaviour of each partner affects that of the other and thus the subsequent course of the relationship. The complexity of this constantly changing relationship can hardly be over-emphasized: the infant's perceptual and motor skills develop, its several responses become directed appropriately and preferentially to the mother, and she likewise comes to respond individually to him. The infant explores his mother and his physical environment, and his cognitive development proceeds as he does so. His mother gradually permits and promotes his independence (see, for example, Harlow & Harlow 1965; Hinde 1974). It will be apparent that, to unravel this complex interaction, it is necessary to establish reliable measures of the behaviour of the participants. Only some of those now used can be mentioned here, but it is convenient to introduce five which show how the mother–infant relationship develops over the first few months of life (figure 1).

Time off (TT off). The proportion of half minutes in which the infant was recorded off its mother.

Relative frequency of rejections ($R/Mk_I + Mk_M + R$). The ratio of the number of occasions on which the infant attempted to make ventro–ventral contact with its mother and was rejected by her, to the total number of occasions it was accepted by her (i.e. contact made on the infant's initiative, Mk_I), picked up by the mother (contact made on mother's initiative, Mk_M), or rejected.

Infant's rôle in ventro–ventral contact (% Mk_I − % Bk_I). The difference between the percentage of the occasions on which ventro–ventral contact was

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made that were due to the infant ($Mk_I\ 100/(Mk_I + Mk_M) = \%\ Mk_I$) minus the percentage of breaks due to the infant ($\%\ Bk_I$).

Time at a distance from mother (> 60 cm). The number of half minutes in which the infant was recorded more than 60 cm from its mother, i.e. out of her reach.

Infant's rôle in maintenance of proximity ($\%\ Ap_I - \%\ L_I$). The difference between the percentage of approaches (distance between mother and infant decreases from more than 60 cm to less) that were due to movement by the infant and the percentage of leavings (distance increases similarly) due to movement of the infant.

It will be noted that Time off and Time out of reach, measured in terms of number of half minutes *in which* the infant was in the conditions stated, will slightly over-estimate the actual times (see, for example, Altmann 1974; Simpson & Simpson 1976). The status of the measures of the infant's rôles in ventro–ventral contact and in proximity have been considered elsewhere (Hinde & Atkinson 1970; Hinde & White 1974).

Figure 1 shows that time off the mother and time at a distance from the mother increase with age of infant. During the early weeks the mother seldom rejects the infant's attempts to gain ventro–ventral contact and is primarily responsible for ventro–ventral contact ($\%\ Mk_I - \%\ Bk_I$ negative). Responsibility for proximity is also primarily maternal or shared ($\%\ Ap_I - \%\ L_I$ near zero). As the infant develops it becomes primarily responsible for nipple contacts (*ca.* week 14) and the proportion of ventro–ventral contacts initiated by the mother diminishes to near zero (*ca.* week 20). Although rejections become frequent, the infant continues to spend *ca.* 30 % of the morning hours on its mother until nearly the end of the first year, though he may be obtaining little milk. Infants are capable of feeding themselves if the mother is removed from week 20 or earlier.

While from this brief account it might appear that, with the increasing frequency of rejections and the appearance of tantrums, the age changes in the relationship involve increasing tension between mother and infant, this is only part of the story. Infant and mother also become progressively more attuned to each other, learning to respond to each other's signals and to align their behaviour with that of the other. For example, in the early weeks a mother is more likely to terminate bouts of ventro–ventral contact initiated by the infant than bouts she had initiated herself, and an infant is more likely to terminate a bout initiated by the mother than a bout he had initiated. This implies that mothers are prone to initiate bouts when the infants are not ready, and the infants when the mothers are not ready. Later the difference disappears, implying that the behaviour of mothers and infants comes to mesh better as the infant develops. Similar generalizations apply to bouts out of ventro–ventral contact (Hinde & White 1974).

This quality of 'meshing' is only one aspect of the mother–infant relationship that requires detailed study. We shall return to this issue shortly.

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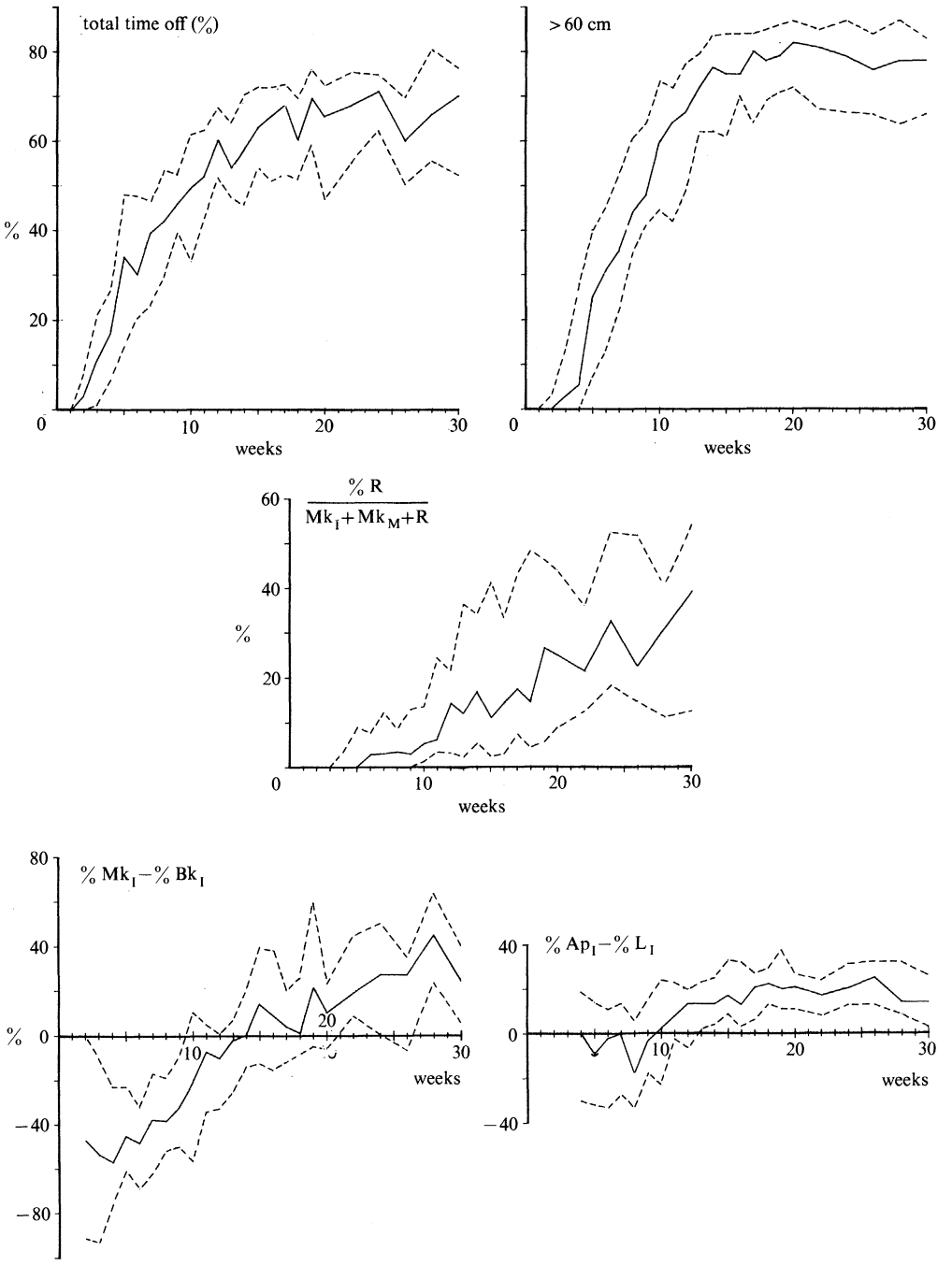


FIGURE 1. For description see opposite.

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Interactions, relationships and social structure

At this point it is necessary to digress briefly. What exactly does one mean by an inter-individual relationship? How can one set about studying the nature of relationships? Such questions demand a conceptual scheme which will underpin the empirical investigations. Such a scheme is presented in figure 2 (Hinde 1976*a*). It involves three levels. *Interactions* are seen as the atoms of social structure. No precise definition of an interaction has been formulated, but the category is intended to include all incidents that can be described in such terms as '*A* does *X* to *B*' or '*A* does *X* to *B* and *B* responds with *Y*'. To describe an interaction, it is necessary to state both what *A* and *B* did together, and how they did it – i.e. the quality of their interaction. For example, *A* might hit *B* gently or forcefully, and *A* might hold *B*'s hand in a restraining or an affectionate manner.

At the molecular level, *relationships* arise when one individual interacts with a known other on a number of occasions. So far as behaviour is concerned, their relationship *is* the nature, quality and patterning of the interactions between them. This is discussed briefly later. Finally *social structure* is seen as the nature, quality and patterning of relationships.

At each of these levels, the investigator must seek for generalizations in two directions. First, from data about the actual interactions that he observes he must abstract generalizations valid over a wider range of instances, contexts or participants. These are indicated in figure 2 by successive blocks of rectangles from left to right. Relationships always involve a degree of abstraction, but further generalizations (necessarily involving greater remoteness from actual data) will be required. The same is true of social structure. Here the pattern of relationships actually observed in a particular instance may be referred to as the 'surface structure', 'structure' being reserved for abstractions of greater generality. In each case the abstractions or generalizations will be in the data language – that is,

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Figure 1. The course of mother–infant interaction in small captive groups of rhesus monkeys. Top left: total time off mother (number of half-minutes in which infant was off mother as proportion of number watched). Top right: time out of arm's reach of mother (number of half-minutes in which infant was more than 60 cm from mother as proportion of number for which it was off her). Centre: relative frequency of rejections (ratio of number of occasions on which infant attempted to gain ventro–ventral contact and was rejected by mother (R) to number of occasions on which it made contact on mother's initiative (Mk_M), made contact on its own initiative (Mk_I), or attempted unsuccessfully to gain contact (R). Bottom left: infant's rôle in ventro–ventral contacts (number of contacts made on infant's initiative, as a percentage of total number made, minus number of contacts broken by infant, as percentage of total number broken). Bottom right: infant's rôle in maintenance of proximity (number of approaches, i.e. distance decreases from > 60 cm to < 60 cm, made by infant, as percentage of total number made, minus number of leavings made by infant, as percentage of total number made). Each point is based on data from 14 to 20 mother–infant pairs. Observations were made between 09h00 and 13h00 over the first 30 weeks of life. Medians and inter-quartile ranges are shown.

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they will employ the same words as were used to describe the basic phenomena originally.

The second direction involves a search for 'organizational principles' which serve to explain the patterning of the data. These are represented by the discontinuous circles in figure 2. They need not be in the data language, and may employ concepts not necessary for the original collection of the data. At the interaction level, these principles will be concerned with the patterning of responses within an interaction and will be primarily physiological, or perhaps psychological, in nature. At the level of the relationship they will be concerned with the patterning of interactions within a relationship: organizational principles concerned with status (including dominance/subordinance), with the effects of interactions on subsequent interactions, and with social forces from outside the relationship, have so far proved valuable. And for the patterning of relationships within the social structure organizational principles concerned with blood relationships,

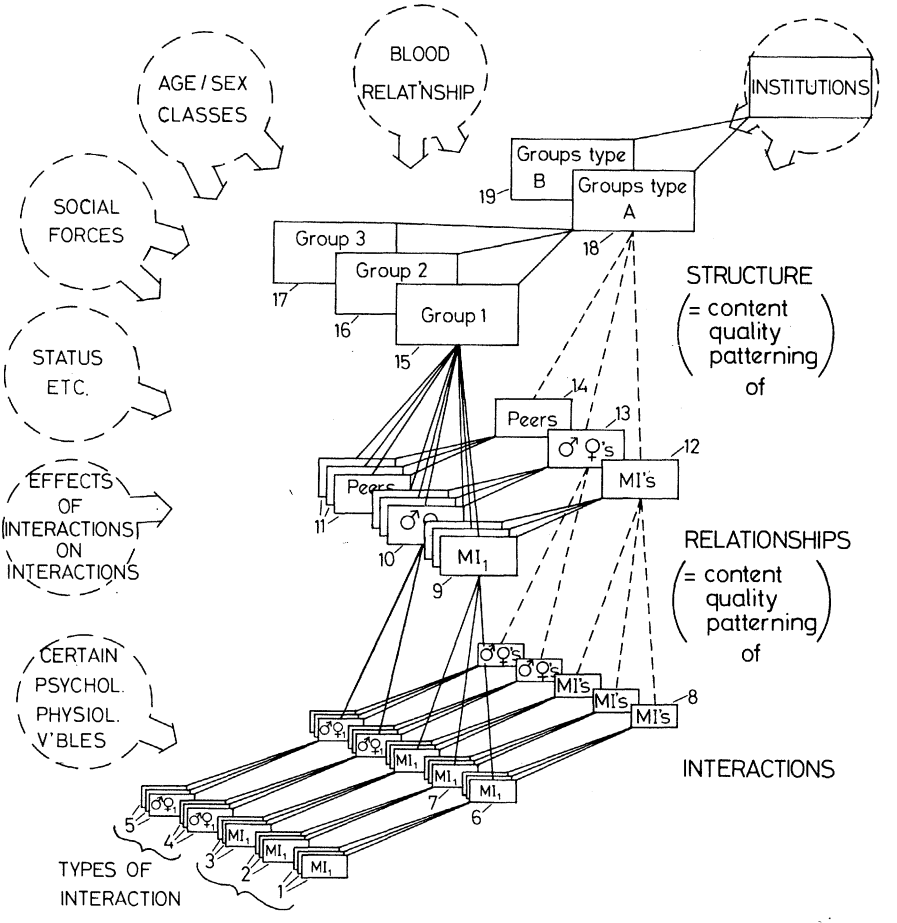


FIGURE 2. For description see opposite.

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relationships within and between age/sex categories, and with status, have proven relevance for non-human primates. It is important to emphasize that these principles will never be expressed in empirical instances in pure form: we shall thus always be concerned with the ways in which they interact.

In the human case a further set of principles operates – those concerned with institutions. The term 'institution' is used here to refer to sets of one or more recognized positions in a society which constrain the behaviour of the incumbents, and thus covers unique positions (e.g. the king), recognized relationships (e.g. marriage) and large-scale systems (e.g. the National Health Service). In that the common feature is one or more positions with recognized rights and duties, it is linked to the concept of rôle.

In fact institutions appear to play a dual rôle in much of the literature of the social sciences. In so far as they constitute goals, patterns of relationships which the

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FIGURE 2. Diagrammatic representation of the relations between interactions, relationships and social structure. Interactions, relationships and social structure are shown as rectangles on three levels, with successive stages of abstraction from left to right. The discontinuous circles represent independent or intervening variables operating at each level. Institutions, having a dual rôle, are shown in both a rectangle and a circle. In the specific instance of a non-human primate, the rectangles might represent:

(1) Instances of grooming interactions between a mother *A* and her infant *B*.
(2) Instances of nursing interactions between *A* and *B*.
(3) Instances of play between *A* and *B*.
(4) Instances of grooming between female *A* and male *C*.
(5) Instances of copulation between *A* and *C*.
(6) First stage abstraction – schematic grooming interactions between *A* and *B*. Abstractions of grooming interactions between other mother–infant pairs are shown behind, but the specific instances from which they were abstracted are not shown.
(7) First stage abstraction – schematic nursing interactions between *A* and *B*. Abstractions of nursing interactions of other mother–infant pairs are shown behind.
(8) Second stage abstraction – schematic grooming interactions between all mother–infant pairs in troop.
(9) Mother–infant relationship between *A* and *B*. Mother–infant relationships of other mother–infant pairs are shown behind (but connections to grooming, nursing, etc., interactions are not shown).
(10) Consort relationship between *A* and *C*. Other consort relationships are shown behind.
(11) Specific relationship of another type (e.g. peer–peer).
(12), (13), (14) Abstraction of mother–infant, consort and peer–peer relationships. These may depend on abstractions of the contributing interactions.
(15) Surface structures of troop containing *A*, *B*, *C*, etc.
(16), (17) Surface structures of other troops (contributing relationships not shown).
(18) Abstraction of structure of troops including that containing *A*, *B*, *C*, etc. This may depend on abstractions of mother–infant, etc., relationships.
(19) Abstraction of structure of a different set of troops (from another environment, species, or etc.).

Rectangles labelled MI_I refer to behaviour of dyad female *A* and her infant *B*. Rectangles labelled ♂♀$_1$ refer to consort pair female *A* and male *C*. Rectangles labelled MIs, ♂♀ refer to generalizations about behaviour of mother–infant dyads and consort pairs respectively (after Hinde 1976*a*).

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incumbents of positions in society strive to achieve, they have the status of organizational principles, and can be used to explain the phenomena observed. But they also appear as abstractions from the data – the best description that an anthropologist can give of, say, the organization of three villages may involve reference to an idealized village council which does not actually exist in any one of them. Institutions are thus shown by both symbols in figure 2.

This conceptual scheme represents perhaps somewhat of a digression from the main theme of this paper, and will not be pursued further here (see Hinde 1976*a*).

The social nexus

At each level in this scheme new properties emerge. Relationships are more than the sum of the interactions of which they are composed in part because the various types of interaction within a relationship may affect each other, and in part because new properties arise which depend on how the interactions are patterned.

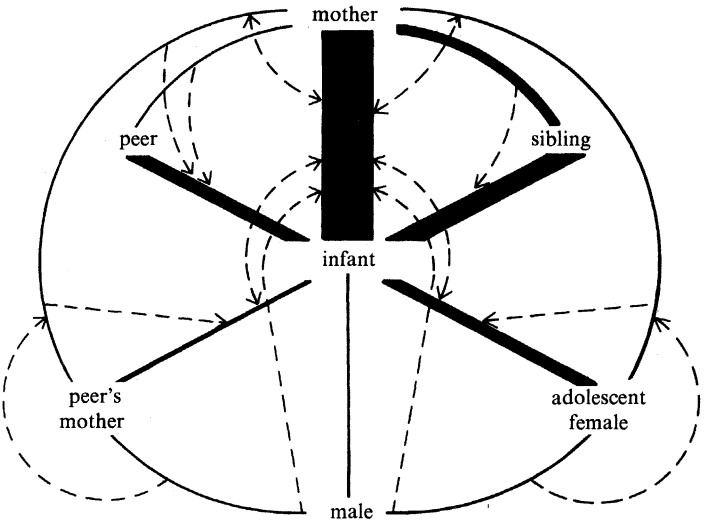


Figure 3. First order zone for a rhesus infant. Continuous lines represent inter-individual relationships, their thickness roughly indicating the importance of the bond. The discontinuous lines show ways in which the relationship has been shown to affect another relationship.

Similarly, at the level of social structure, each dyadic relationship within a group is affected by the other relationships within that group. Two examples of the latter from our rhesus monkey groups may be mentioned.

(i) The mother–infant relationship and social companions. The extent to which a mother permits or promotes the independence of her infant is affected by the presence of social companions. Infants of isolated mother–infant pairs spend more time off their mothers and more of that time at a distance from their mothers,

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than do infants of group-living mothers. This is correlated with a greater frequency of rejections on the part of their mothers and an increased rôle of the infants in maintaining proximity to their mothers while off her (Hinde & Spencer-Booth 1967). These correlations suggest that the differences in the relationship are due primarily to differences between mothers (see below).

(ii) The infant's social nexus. Figure 3 is a qualitative representation of the types of relationship which affect an infant, the thickness of the continuous lines indicating the 'strength' of its relationships with its various social companions and of their relationships with each other. The discontinuous lines represent the effects of one relationship on another. For instance, adolescent females are prone to attempt to cuddle, carry and groom young infants, but the extent to which they are permitted to do so by the mother depends on the nature of the relationship between the infant's mother and the adolescent. That relationship in its turn may be affected by the male's relationship with the adolescent; if she is his favourite, he may intervene on her behalf in a dispute with the infant's mother. While figure 2 is to be interpreted in purely qualitative terms, each of the lines drawn is based on quantitative data, and it does represent the complexity of the effects of relationships upon relationships within even a small group.

The effects of brief periods of mother–infant separation

We are now in a position to consider data derived from experiments involving brief periods of separation between mother and infant. These experiments were conducted when the infants were between 21 and 32 weeks old, and thus quite able to feed themselves. The earlier experiments involved the removal of the mother from the pen for 6 or 13 days, at the end of which time she was returned to the group (Hinde & Spencer-Booth 1971*a*, and references cited therein). The procedures and measures used are explained in the figure legends.

We may consider first the general nature of the infant's response (see figures 4, 5 and 6, data labelled M/R). When the mother is removed there is a great increase in 'whoo' distress calling. This calling decreases after the first day, but is likely to remain higher than before separation for some weeks after reunion. The infant frequently sits in a hunched depressed posture, and at the same time its locomotor activity falls. Upon reunion, some, but not all, infants spend a great deal of time on their mothers for the first few days. At this time tantrums are frequent. During the weeks after reunion there is often a complex interaction between mother and infant, with the infant being very demanding of its mother's attentions, the mother at first yielding to its demands, then becoming intolerant, and so on until a more stable relationship is again established.

Although the course of events always followed this general pattern, there were marked individual differences in the extent and severity of the effects. In searching for factors related to these individual differences, we used primarily the four

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measures of infant behaviour when off its mother – locomotor activity in terms of number of sections of cage entered, activity when active (a measure of the proportion of the period in which the infant was active at all in which it was hyperactive), sitting still and adoption of the hunched posture, and whoo-calling. However, since some infants tended to be affected in one way and others in another, we also calculated an overall rank based on the sum of the inverse rank of locomotor activity and the actual ranks on sitting hunched and whoo calling. This 'distress index' provided only a crude overall measure of the behaviour exhibited, but was useful in this context.

First, the length of the separation period proved to be an important variable. Infants which had been separated for 13 days showed a greater and more prolonged depression of locomotor activity, and a higher rate of distress calling, than did those which had been separated for one or two periods of 6 days.

Of other factors that could affect the response to separation, we found that the age of the infant (within the range studied) and the extent to which the infant was taken on by other group companions during its mother's absence, were at most of minor importance in our groups. Since nearly all mothers were multiparous and most infants had peers present in the cage, the influences of these factors could not be assessed. There were, however, some differences related to the sex of the infant: during separation and after reunion male infants whoo-called more than did females.

A more important issue was the relationship between individual differences before and after separation. Although the median scores for locomotion fell, and for sitting hunched and for whoo-calling rose, as a result of separation, the rank order correlation coefficients between pre- and post-separation scores were mostly significantly positive. Thus the effects of the mother's removal can be seen as an accentuation of pre-existing trends.

However, an even more crucial issue turned out to be the nature of the mother–infant relationship. Table 1 shows the rank order correlation coefficients between the distress index and four measures of the mother–infant relationship before separation. There was little relationship with the time that the infant had spent on or near the mother, but those infants that were most distressed in the two later post-reunion periods tended to be those that had been rejected most by their mothers and that had had to play the greatest relative rôle in staying near them, when off them. The distress index during these post-reunion periods was related also to contemporaneous values of these measures of the mother–infant relationship: examination of Kendall tau partial correlation coefficients showed that the relationship of the distress index to the pre-separation measures of the mother relationship was strongest at reunion and wanes with time, while that to the post-reunion measures of the relationship increased with time.

Thus the experiments up to this point demonstrated that a separation experience could produce symptoms in rhesus monkeys which are at least comparable with those seen in man, that the response can in part be regarded as an exacerbation of

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pre-existing tendencies, and that it is closely related to the nature of the mother–infant relationship. The next problem that concerned us was, what is it precisely about a separation experience that produces these effects? Is it separation from the mother *per se*; or the associated emotional deprivation (Howells 1970); or the stresses that almost inevitably accompany a separation experience (Robertson & Robertson 1971), or the long-term disruption of the ongoing mother–infant

TABLE 1. SPEARMAN RANK ORDER CORRELATION COEFFICIENTS BETWEEN DISTRESS INDEX IN THREE POST-SEPARATION PERIODS AND PRESEPARATION MEASURES OF MOTHER–INFANT RELATIONSHIP

(Day 10 was the day of reunion.)

measures of mother–infant relationship	correlation coefficient		
	day 10	days 11–16	days 23–27
time off	+0.02	−0.07	−0.13
$R/(Mk_I + Mk_M + R)$	+0.41	+0.59*	+0.43*
> 60 cm	−0.26	−0.28	−0.18
$\%\ Ap_I - \%L_I$	+0.30	+0.64**	+0.41

* $P < 0.05$, ** $P < 0.01$.

TABLE 2. FOUR SEPARATION PROCEDURES USED TO STUDY EFFECTS OF TEMPORARY SEPARATION OF INFANT FROM MOTHER RHESUS MONKEY

(Separation and/or removal occurred on day 4 and reunion on day 17.)

	N	mother's pen	infant's pen	separation
M/R	6	strange	home	yes
I/R	6	home	strange	yes
M−I/R	7	strange	strange	yes
(MI)/R	7	strange	strange	no

relationship, or what? In attempts to make some progress with these problems, the separation procedure was varied. Whereas in the experiments discussed so far the mother was removed from the home pen, the infant staying in the physical and social environment to which it was accustomed, in later experiments either the infant, or both infant and mother, were removed from the pen: in the latter case infant and mother were either confined together or separated. Since all separations lasted thirteen days, the data can be compared with those from the 13 day separations mentioned above. The groups concerned are shown in table 2. Some of these experiments were conducted by Linda McGinnis (1975; Hinde & Davies 1972; Hinde & McGinnis, in the press).

There were no significant intergroup differences in measures of the mother–infant relationship before separation, though infants in the groups in which only the mother or only the infant were removed from the home pen scored more highly on activity measures before separation.

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In considering the data on the effects of separation, three points must be borne in mind.

(i) Activity scores during the separation period for groups in which the infant was removed from the home pen and confined in a small cage are not comparable with those in which the mother was removed and the infant remained in the $5\frac{1}{2}$ m long home pen.

(ii) In any case activity scores are interpretable only in relation to other measures, for a high activity score can result from relaxed play and exploration, or from frantic searching for an absent mother.

(iii) As will become apparent later, a crucial issue in determining the post-reunion behaviour of the infants was the mother's preoccupation with re-establishing relationships with her group companions.

Considering first the behaviour of the infants during the separation period (figure 4), those infants who remained in their home pens (M/R) showed moderate locomotor activity and high distress calling on the day of separation, but soon became depressed, with locomotor activity falling to a low level. Their rate of whoo-calling was initially high but soon fell. The proportion of time sitting hunched, though falling a little, remained at a fairly high level.

Turning to the groups in which the infants were moved to a strange pen, the infants which remained with their mothers ((MI)/R) showed on the first day a marked decrease in locomotor activity which recovered quickly to levels comparable with those shown in the larger pens before separation, but remained much lower than either of the other infant-removed groups. The infants were in fact relatively little disturbed.

Data on the other two infant-removed groups (I/R and M – I/R) for the separation experience can be considered together since, in both cases, the infant was separated from its mother and placed in a strange cage. There was, however, a procedural difference in the way that the data were collected: with the group in which the mothers were left in the home pen the observer was in sight of the infants, while with the group in which the mothers were also removed and placed elsewhere the observer was in a hide behind one-way glass. Since the infants were well habituated to observers, we do not believe that the procedure adopted in the first case made much difference to their behaviour.

In both these infant-removed and separated groups the infants showed high activity levels, with the separation means even higher than the levels for the day of separation. (Two males in the M – I/R group were almost immobile with fear on the first day, and this partially accounts for the difference between the day of separation and the rest of the separation period.) Their activity consisted in part of bouts of frantic pacing, as indicated by their high scores on activity when active ($> 5/> 1$, figures 4, 5). Between such bouts they sat immobile, but depression was less evident in these groups than amongst the separated infants who remained in their home pens. The rate of whoo-calling in both infant-removed-and-separated groups (I/R and M – I/R) was high during the separation period.

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From these data we can conclude that the response to separation runs a different course according to whether the infant is kept in a strange small cage or the familiar home pen during the separation period. In the latter case symptoms which can be likened to those of depression set in fairly soon after separation, whereas if the infant is in a strange small cage 'searching' behaviour persists for longer. Furthermore infants removed to a strange small cage but accompanied by their mothers

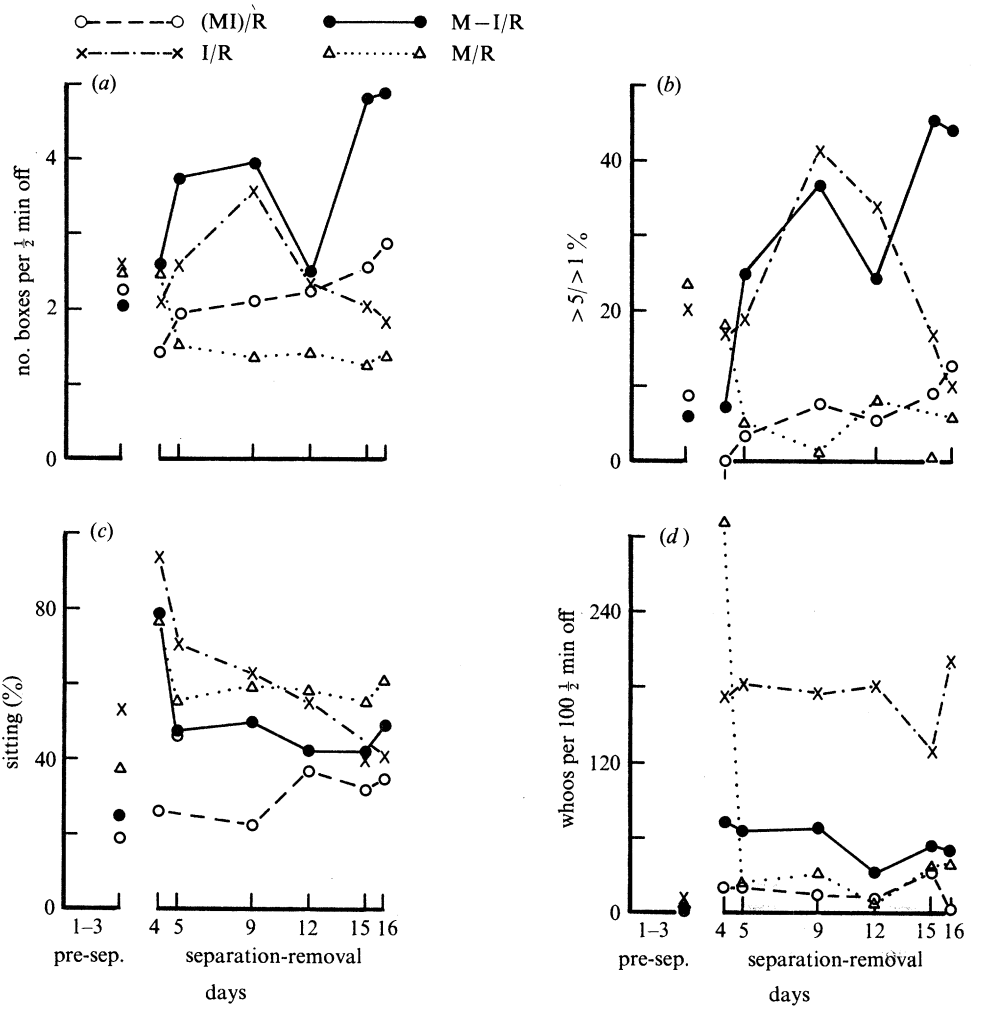


FIGURE 4. Behaviour of infants before and during separation/removal – group medians. Separation and/or removal started on day 4, and reunion was on day 17. No. of boxes per half a minute off is a measure of locomotor activity. > 5/> 1 % is a measure of how active the infants were when they were active – specifically, the number of half minutes in which they entered more than 5 of the 16 boxes, into which the pen was divided, divided by the number of half minutes in which they entered more than one. Sitting %: number of half minutes in which they sat in the hunched depressed posture; Whoos per 100 ½ min off is a measure of the actual frequency of distress calls.

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show relatively little disturbance. Separation from the mother is thus a crucial issue.

We may now consider infant behaviour after reunion ((MI)/R) (figure 5). The group in which mother and infant were not separated showed an increase in activity on reunion: this activity consisted in large part of play with their peers, which continued at a high level for several weeks. These infants showed at first little whoo-calling, though this did increase to moderate levels later. By and large these infants were little disturbed.

With the infants whose mothers were moved (M/R), the depression-like symptoms which set in early in the separation period persisted after reunion: for

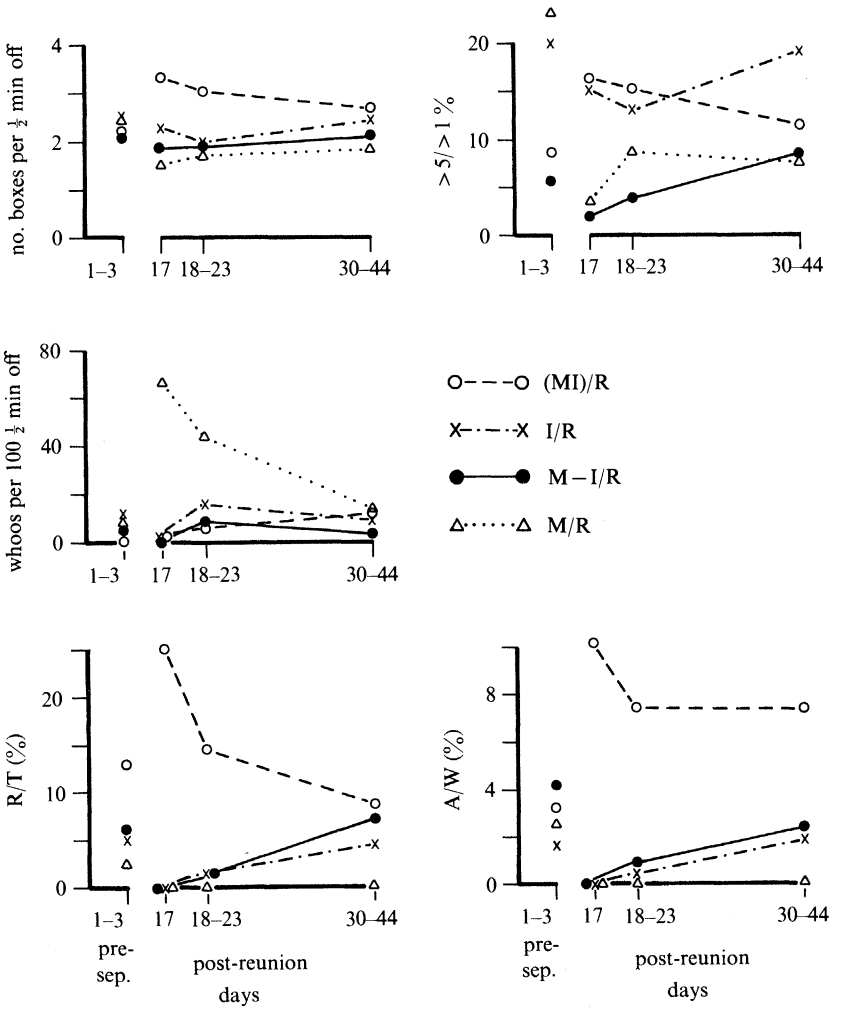


FIGURE 5. Behaviour of infants before separation/removal and after reunion – group medians. R/T (%) and A/W (%) are measures of the frequency of rough-and-tumble play and approach/withdrawal play respectively. Other measures as in figure 4.

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some weeks activity levels remained lower, and sitting and whoo-calling levels higher, than before separation. In these respects they differed from the infants who were separated by removal to a strange pen (I/R and M – I/R): in these the median activity levels approached more closely their pre-separation values, and sitting hunched and whoo-calling were lower, than with the mother-removed infants.

These differences can be related to the mother–infant relationship on reunion (figure 6). Consider first the infant removed (I/R) and mother-removed (M/R) groups. Within the latter, the frequency of rejections and the infant's rôle in ventro–ventral contact were high on reunion. With the infant-removed group, the reverse was the case. Thus the greater distress shown by the mother-removed group can be understood as a consequence of the less adequate maternal care they received on reunion. We believe that these differences arose because mothers removed and then re-introduced to the pen were preoccupied with re-establishing relationships with their group companions, and were therefore less attentive to their infants. Some data on this were analysed extensively by L. McGinnis, and showed that the effect was especially marked with the less sociable, low-ranking mothers. By contrast, in the group in which only the infants were removed, the mothers staying with their group companions in the home pen, the mothers could devote all their energies on the day of reunion towards meeting the demands of their infants.

We may now consider the group in which infants and mothers were removed and separated from each other (M – I/R). These mothers also had to re-establish their relationships with their group companions. The frequency of rejections, and the relative rôles of infants in contact and proximity, were again high. However, unlike the infants of the M/R group, these M – I/R infants showed little sign of depression. Their activity levels were little below pre-separation levels, and their play behaviour showed considerable recovery. It was probably for this reason that their level of distress calling was much lower than that of the M/R group.

Mothers of the non-separated (MI)/R group also had to re-establish their relationships with their group companions, but here no separation had occurred. In this group, the frequency of rejections, and the infants rôle in contact and proximity, were all relatively low. Presumably the more receptive behaviour of the mothers in this case was related to the fact that the mother–infant relationships had not been interrupted.

In summarizing these findings, four issues may be emphasized.

(1) Infants who remain in their home pen while their mothers are removed become depressed and remain so after reunion. The persistence of the depression could be due to a persisting physiological state or to conditioning to the environment in which separation occurred, or both.

(2) Mothers which are removed from the home pen must re-establish their relationships with their group companions on return, and thus have less to spare for their infants.

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(3) Depressed infants are less effective in eliciting maternal care than are non-depressed infants.

(4) If the mother–infant relationship is interrupted by separation, mothers are less prone to be adequately maternal on reunion, than they would be if they had been continuously together.

Thus in the M/R group the infants remained depressed, although they called

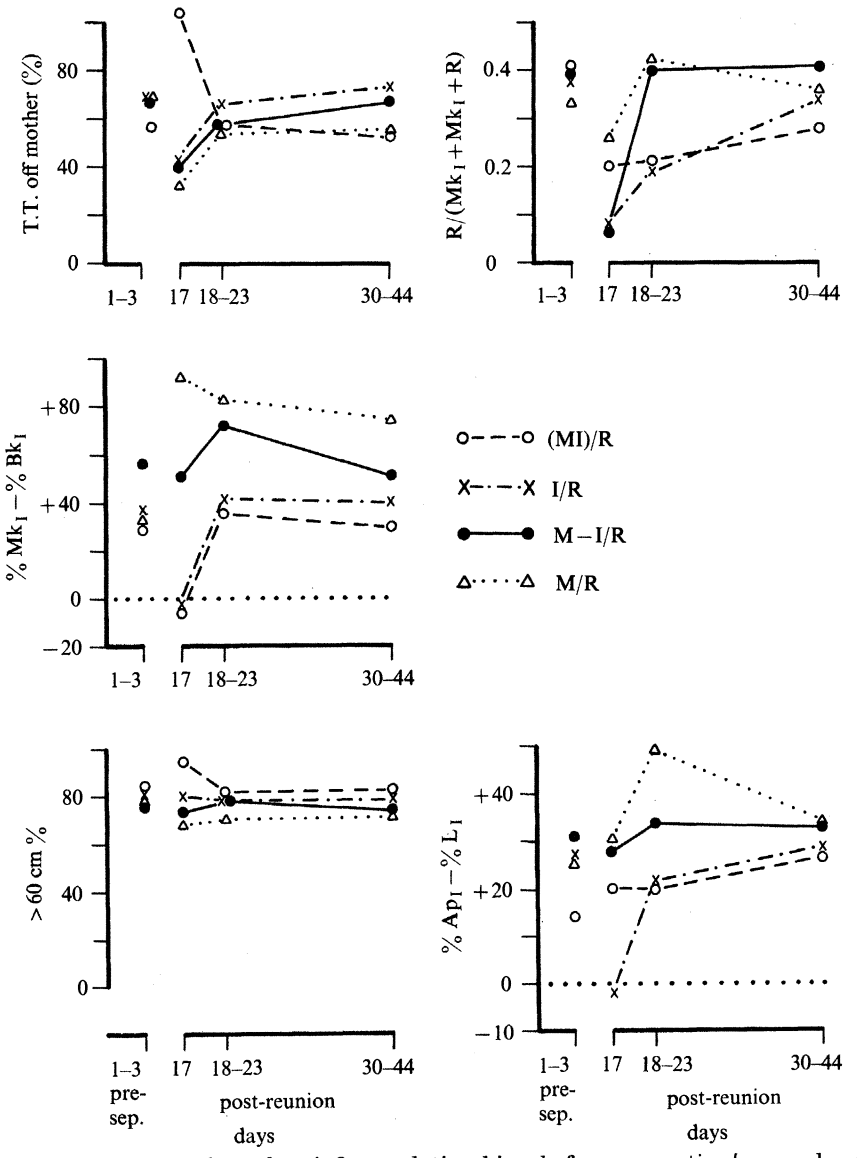


FIGURE 6. Measures of mother–infant relationships before separation/removal and after reunion – group medians. The measures used are described in figure 1.

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frequently they did not pursue their mothers, the mothers were distracted by their other group companions, and the infants were not sufficiently persistent to direct their mother's attentions to themselves.

In the M – I/R group the situation was similar but, since the infants had not been in the home pen, they were not so depressed and were more persistent in pursuing their mothers. The high activity levels shown by this group can be partially understood in these terms.

Infants of the infant-removed group were not only not depressed, but also had fully responsive mothers. They were thus less affected than infants in the other separated groups.

In the group where both mother and infant were removed and not separated the infants were again not depressed, and the mothers were responsive to them presumably because their relationship had been unbroken. On reunion both mothers and infants were concerned with re-establishing relationships with group companions and the infants at first spent much time off their mothers, while keeping visual contact with them.

One last aspect of these separation experiments must be discussed. So far we have been concerned only with the consequences of the separation experience in the immediately ensuing weeks. While these can be profound, it remains possible that in the longer term all effects of the separation experience are erased.

Some of our previously separated animals have become old enough for us to assess this. We have gathered data on their behaviour when 1 year, 18 months and 2½ years of age. So far data are available only for infants subjected to one or two 6 day separation experiences.

In brief we could detect no clear differences between these infants and non-separated controls when we observed them in their home pens. When tested with strange objects in their home pens, some suggestive differences emerged. But when tested with strange objects in a strange environment clear differences appeared: these differences can be summarized by saying that the previously separated infants were less able to cope with slightly alarming situations (Spencer-Booth & Hinde 1971).

The nature of the mother–infant relationship

These experiments have produced clear evidence that the consequences of a separation experience are related to the nature of the mother–infant relationship. They indicated that, in the conditions of these experiments and within the range exhibited by our subjects, the amount of time that infant and mother spent in contact with or near each other were of at most minor importance. But certain measures which can be described as measures of tension in the mother–infant relationship – namely the relative frequency of rejections and the infant's rôle in the maintenance of mutual proximity – are correlated with the effects of the separation experience on the infant.

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This of course raises questions about the development of the mother–infant relationship. What determines the individual characteristics of each mother–infant relationship?

We have recently completed an analysis of 63 mother–infant pairs, in an attempt to obtain some information on this issue. When divided according to sex of infant, availability of peers, and dominance status, parity and source of mother, relatively few subjects fell into each subgroup. However it was possible to make a number of comparisons for each of the factors considered (e.g. male versus female infant; dominant versus subordinate mother). These showed that although all the variables studied, except perhaps maternal parity, affected the relationship, none produced effects that were wholly consistent over all comparisons. Interactions were thus ubiquitous, and we must await even more extensive data before the various factors involved can be disentangled (White & Hinde 1975).

The experiments reviewed above show also that the consequences of a separation experience depend in part on the extent to which the relationship is disturbed as a consequence of the separation. Further progress here demands more basic knowledge about the nature and dynamics of inter-individual relationships than is at present available. At present, much of what is known depends either on case history studies, with general principles relatively inaccessible, or on investigations, many excellent in themselves, but designed to illustrate particular points of view (Swenson 1973). Although the nature and dynamics of human interpersonal relationships pose problems even more complex than those with which biologists usually deal, it seems possible that biology can make some contribution, for two reasons.

First, if there is to be a science of interpersonal relationships, it must surely rest on a descriptive base. Biologists, whose science depends on the work of generations of taxonomists and systematists, may be able to contribute towards providing this. As a start, certain dimensions along which interpersonal relationship differ, and which may be important for understanding their dynamics, have been specified (Hinde 1976*b*). But it is crucial to be able to specify the qualities of interpersonal relationships in reasonably objective terms. Here the biologist may contribute by using relatively simple animal cases to work out methods. Consider for example, the extent to which the behaviour of each partner in the relationship is directed with respect to the ongoing goals of the other. We have already seen that this 'meshing' as assessed in one way, increases with age. Using a slightly different measure, M. J. A. Simpson has shown that at any one age rhesus monkey mother–infant pairs differ in the extent to which the behaviour of each partner 'meshes' with that of the other. Figure 7 contains data on two dyads. It shows the probability that each partner will leave the other in successive time intervals after a particular reference event – namely an approach by the infant to the mother. In one dyad the probabilities that mother and infant will leave run closely parallel, whilst in the other they bear little relationship to each other. In the former the mother is likely to leave when the infant is likely to leave, and vice versa: we could

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describe them as, by this criterion, a 'well-meshed' pair (Hinde & Simpson 1975); and it is at least possible that 'meshing' is a property akin to 'intersubjectivity' (Asch 1952).

The second way in which biologists may contribute also stems from the relative simplicity of animal inter-individual relationships. It has become apparent that,

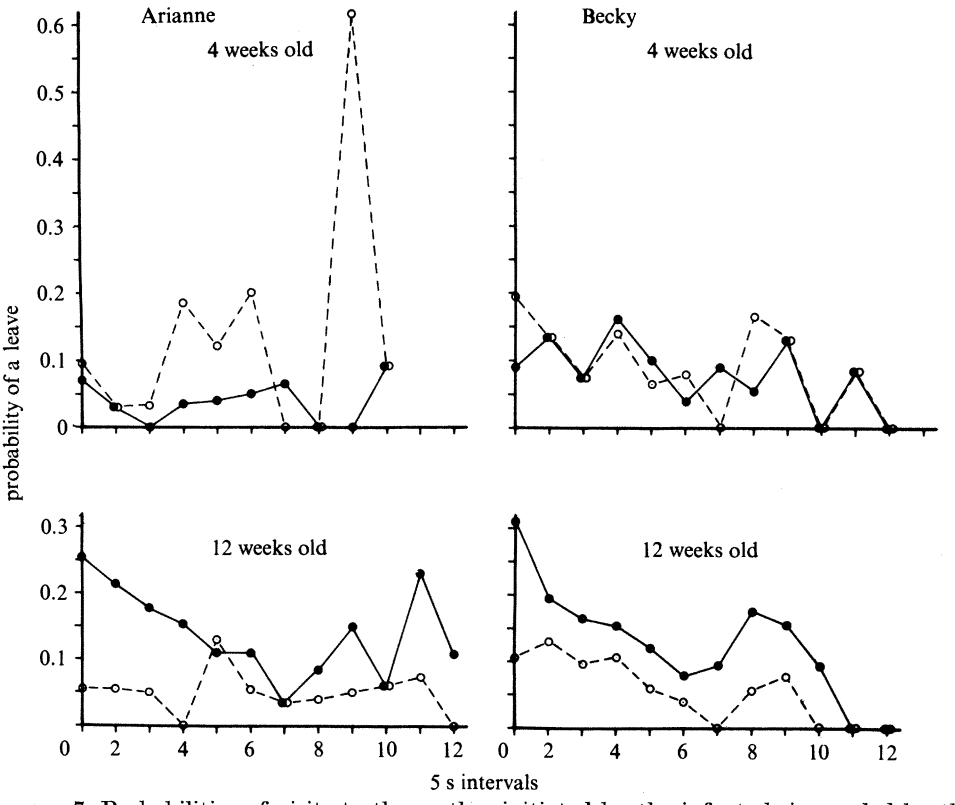


FIGURE 7. Probabilities of visits to the mother initiated by the infants being ended by the mother (○) and by the infant (●) when the visits had lasted between 0 and 5 s, 5 and 10 s, 10 and 15 s, and so on (from Hinde & Simpson 1975).

in studying relationships, it is extremely difficult to obtain measures of the behaviour or of the behavioural propensities of either partner within the relationship that are independent of the behaviour of the other. For instance, how quickly a mother responds to her baby's cry, which might seem to be a measure of maternal behaviour, must surely reflect also how often the baby has cried recently, itself in turn perhaps dependent on maternal responsiveness. For this reason, the contributions of the two partners to the nature of the relationship at any one time, to changes in its nature with time, or to differences between relationships, are difficult to disentangle. However studies of rhesus monkeys have shown that some progress can be made by using correlations between measures. The method has

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been discussed elsewhere (see, for example, Hinde 1969, 1975; Hinde & Herrmann 1976) and one example will suffice. Suppose one infant spends more time off its mother than another, is the difference primarily due to a difference between mothers or to a difference between infants? If attempts by the former infant to make contact are more frequently rejected than attempts by the latter, the difference in time off must be due primarily to a difference between the mothers, but if it is off more even though it is rejected less, the difference must be primarily between the infants. Similarly if, for a number of dyads, rank order on time off is positively correlated with rank order on frequency of rejections at a particular age, inter-mother differences are primarily responsible for inter-dyad differences in time off. This method, basically very simple, can be used to answer questions of the types posed above and thereby to tease apart the dynamics of the relationship. Further progress must of course depend on the specification of principles that will provide understanding of the dynamics of inter-individual relationships – that is of how interactions affect subsequent interactions, and of how relationships may change their properties with time, yet preserve their integrity in spite of changes in the nature of the individuals concerned and the vicissitudes of their respective social worlds (see, for example, figure 2 and accompanying discussion). Some of the issues involved have been discussed elsewhere (Hinde & Stevenson-Hinde 1976) but two points arising from these separation experiments may be mentioned here.

(i) *Continuous or threshold effects*? The finding that any particular factor affects the course of a relationship proves neither that the presence of that factor *in any degree* is of importance, nor that there is a straight line relationship between the extent to which it is present and its effects. For example, these experiments show that length of separation is an important issue. However, common sense indicates that very short separation experiences will have no or negligible consequences. Furthermore, while marked differences between the effects of 1 and 20 day separation experiences are to be expected, effects of 30 and 50 day separations might be quite similar. In other words we must expect a threshold region, below which a separation experience has at most minor consequences, and above which they are major. Of course the threshold need not be sharply defined.

(ii) *Regulation versus positive feedback*. That relatively minor perturbations produce negligible long-term effects on the course of the mother–infant relationship implies that it has considerable capacity for regulation (cf. Dunn 1976). Where marked diversions do occur, they can sometimes be understood in terms of positive feedback, the consequences of a minor deviation becoming magnified and over-riding the regulatory mechanisms. An example is perhaps provided by the infants left in the home pen while their mothers were removed (M/R): after reunion their continued high level of distress calling was a consequence of two factors, the mothers' social distractions (present also in the M – I/R group) and the infants' depression, the former exacerbating the latter and leading to a more permanent distortion of the relationship which was not fully corrected, at least in the weeks following reunion.

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Conclusion

These experiments were started with reference to a particular controversy – can a period of separation from the mother affect the personality development of the human infant? The evidence that a brief period of separation can have long-term effects in monkeys adds force to the evidence, now itself considerable, that long term effects *can* occur in man (see, for example, Bowlby 1969, 1973; Rutter 1972; Douglas 1975). Evidence comes not only from the experiments reviewed here, but also from those conducted in a number of other laboratories (see Suomi 1974; Kaufman 1970). The monkey data also provide evidence about what sort of individuals are most likely to be most affected, and how the effects can best be ameliorated – evidence whose relevance to the human case is certainly worth assessing. But this material also opens up questions of even more far-reaching importance – questions concerning the nature and dynamics of inter-individual relationships. Of course monkeys are different from people, and of course human personal relationships are infinitely more complex than relationships between monkeys. But just because monkeys are *relatively* simple, their study may the more readily be used to harden up concepts, to generate principles and to suggest methods which are worth trying out in the human case.

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